TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 4: Professional Bodies and Codes

Obligations to Society, Employers & Profession, Engineering Contract Law, Limitations of Codes & Fundamental Canons

PART 4

Professional bodies and codes

3 sections, about 1 h 55 min of reading.

Model rules: obligations to society, employers and engineers

~35 min to read | High priority | 6 questions

Source: Set 4, slides 11-25 (UIPE and ERB Model Rules of Professional Conduct)

The model rules and their preamble

Slide 11 names the two Ugandan bodies whose model rules the lecturer uses. **UIPE** is the **Uganda Institute of Professional Engineers**. **ERB** is the **Engineers' Registration Board**. Their *Model Rules of Professional Conduct* say what a registered engineer must and must not do. The rules are grouped by the party the engineer owes the duty to. There are three groups, and a past question asks for exactly these three:

1. Obligation to **society** (slides 13-15).
2. Obligation to **employers and clients** (slides 16-22).
3. Obligation to **other engineers**, which the exam calls "the engineering profession" (slides 23-25).

The preamble (slide 12)

The preamble states the purpose of the whole set of rules: to **safeguard life, health, and property**, **to promote the public welfare**, and **to maintain a high standard of integrity and practice**. Every later rule serves one of these three aims. When you answer an obligations question, you can open with this sentence and then name the three groups.

> **Key point.** The slides number the rules separately inside each group (1, 2, 3 ... in society; 1, 2 ... in employers). Do not try to memorise a single rule number. Memorise the heading words, such as "whistle blowing", "clean hands" and "full disclosure", because those are the labels the lecturer gave.

> **Watch out.** The past solution files answer "obligations to each stakeholder" with generic points (environmental impact, continuous learning, mentorship, community engagement). Those points are not in the slides. The slides contain specific rules, and the model answers below follow the slides. Where a solution-file point is useful but not on the slides, it is marked "beyond the slides".

Obligation to society

Slides 13-15 list eight rules under the heading "Obligation to Society". The slide itself gives the group a sub-heading for each cluster. The wording below is the slide wording.

Broad context of responsibility (rules 1 and 2)

1. "While performing services, the engineer's **foremost responsibility is to the public welfare**."
2. "Engineers shall **approve only those designs that safeguard the life, health, welfare, and property of the public** while conforming to accepted engineering standards."

Rule 1 sets the order of loyalty: public first, employer and client second. Rule 2 turns that into a test at the approval stage. An engineer signs off a design only if it protects the public and meets accepted standards. This is why a seal or signature carries ethical weight.

Whistle blowing (rule 3)

3. "If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall **notify his/her employer or client and any appropriate authority**."

Read the trigger carefully. It has two parts: the engineer's professional judgment was *overruled*, and the result is *danger* to the public. Only then must the engineer notify two parties: first the employer or client, and also any appropriate authority. The slide does not say "go to the press". The word is "authority". For the theory of whistleblowing (conditions, internal versus external, effects on the whistleblower) see section t21.

Truth in duties (rules 4 and 5)

4. "Engineers shall be **objective and truthful** in professional reports, statements, or testimonies and shall **provide all pertinent supporting information** relating to such items."
5. "Engineers shall not express a professional opinion publicly unless it is **based upon knowledge of the facts and a competent evaluation of the subject matter**."

Rule 4 covers what the engineer writes or says in an official capacity: reports, statements and testimonies (for example in court). Two duties: be objective and truthful, and supply the supporting information, so nothing relevant is left out. Rule 5 covers public opinions. A public opinion is allowed only when two conditions are met: knowledge of the facts, and a competent evaluation.

The duty of full disclosure (rule 6)

6. "Engineers shall not express professional opinion on subject matters for which they are motivated or paid, unless they **explicitly identify the parties on whose behalf** they are expressing the opinion and **reveal the parties' interest in the matters**."

This is about hidden sponsors. If you are paid or motivated to speak on a matter, the audience must be told on whose behalf you speak and what that party stands to gain. Example (beyond the slides): an engineer paid by a cement company who praises that company's product on radio must say that the company is paying.

The "clean hands" rule (rule 7)

7. "Engineers shall not enter business ventures or permit their names or their firm's names to be used by any persons or firm which is **engaging in dishonest, fraudulent, or illegal business practice**."

The engineer keeps clean hands in two ways: by not joining such ventures, and by not lending a personal or firm name to them. A respected name gives a dishonest business credibility, and the rule forbids that.

Final obligation to society (rule 8)

8. "Engineers who have knowledge of possible violation of any of the rules listed in this and the following two parts shall **provide pertinent information and assist the state board** in reaching final determination of the possible violation."

"This and the following two parts" means society, employers and clients, and other engineers. If you know of a possible breach of any of them, you must give the information and help the board decide. In Uganda the board is the ERB (the slide says "state board", from the model it was taken from). This makes the profession self-policing.

No.	Slide heading	One-line meaning
1	Broad context of responsibility	Public welfare comes first
2	Broad context of responsibility	Approve only safe designs that meet standards
3	Whistle blowing	If overruled and public is in danger, notify employer or client and authority
4	Truth in duties	Objective, truthful, with all supporting information
5	Truth in duties	Public opinion only on facts and competent evaluation
6	Duty of full disclosure	Name the party you speak for and its interest
7	"Clean hands" rule	No ventures or name-lending with dishonest, fraudulent or illegal business
8	Final obligation to society	Report possible violations and help the board

Q: **QB B4(i) · 3 marks** **Describe a registered engineer's obligation to the following stakeholder group: (i) Society.**

MARKER WORDS: OBLIGATION TO SOCIETY

Say it like this:

> An engineer's **foremost responsibility is to the public welfare**: to **safeguard life, health, and property**, to approve only designs that conform to **accepted engineering standards**, and to **notify the employer or client and any appropriate authority** when overruled in a way that endangers the public.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
public welfare	saying who comes first when duties clash	**SHOULD**
life, health, welfare and property	describing what designs must safeguard	**MUST**
objective and truthful	describing duties in reports, statements, testimony	**MUST**
competent evaluation	explaining when a public opinion is allowed	**SHOULD**
whistle blowing	explaining what to do when judgment is overruled	**MUST**
appropriate authority	saying who must be notified besides the employer	**SHOULD**
full disclosure	explaining paid or motivated opinions	**SHOULD**
clean hands	explaining why names must not be lent to dishonest firms	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Purpose of the model rules (the preamble).
To safeguard life, health, and property, to promote the public welfare, and to maintain a high standard of integrity and practice.

The three groups the model rules place obligations towards.
Society; employers and clients; other engineers.

Whistle blowing rule (society, rule 3).
If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall notify his/her employer or client and any appropriate authority.

The slide headings for the eight obligations to society.
Broad context of responsibility (1, 2); whistle blowing (3); truth in duties (4, 5); the duty of full disclosure (6); "clean hands" rule (7); final obligation to society (8).

The "clean hands" rule.
Engineers shall not enter business ventures or permit their names or their firm's names to be used by any persons or firm which is engaging in dishonest, fraudulent, or illegal business practice.

The final obligation to society.
Engineers who have knowledge of possible violation of any of the rules shall provide pertinent information and assist the state board in reaching final determination of the possible violation.

Obligation to employers and clients

Slides 16-22 give the engineer's duties to the people who hire or pay them. Slides 16 and 17 are the same slide repeated, so there are six headings: professional competence, validity of approvals, confidentiality, conflict of interest, full disclosure, and government conflicts of interest.

Professional competence (slides 16-17)

1. "Engineers shall **not undertake technical assignments for which they are not qualified**."
2. "Engineers shall **approve or seal only those plans or designs** that deal with subjects in which they are competent and which have been **prepared under their direct control and supervision**."

Rule 1 stops an engineer accepting work outside their training. Rule 2 is about the seal. Two conditions must both hold before sealing: the subject is within the engineer's competence, and the work was prepared under the engineer's direct control and supervision. Sealing a junior's drawings without supervising them breaks the rule. (This links to the individual competence principle in section t04.)

The validity of approvals (slide 18)

"Engineers may coordinate an entire project provided that **each design component is signed or sealed by the engineer responsible** for that design component."

A project manager-engineer may lead a big project made of many disciplines. The project is valid only when each part carries the signature or seal of the engineer who is responsible for that part. The coordinator does not seal what they are not competent in. Example (beyond the slides): on a building, the structural engineer seals the structure and the electrical engineer seals the wiring, and the coordinator seals neither on their behalf.

Confidentiality requirement (slide 19)

"Engineers shall **not reveal professional information without the employer's or client's prior consent** except as authorized or required by law."

Two points. The default is silence, and the way out is the *prior* consent of the employer or client, not consent given afterwards. The one exception is where the law authorizes or requires disclosure, for example a court order.

Conflict of interest (slide 20)

1. "Engineers shall **not solicit or accept direct or indirect considerations, financial or otherwise**, from contractors, their agents, or other parties while performing work for employers or clients."
2. "Engineers shall **disclose to their employers or clients potential conflicts of interest** or any other circumstances that could influence or appear to influence their professional judgment or their service quality."

Rule 1 bans taking benefits (gifts, money, favours) from contractors or others while working for an employer or client. "Direct or indirect" closes the loophole of benefits passed through a relative or agent. Rule 2 is a duty to disclose. Note the words "potential" and "appear to influence": the engineer must disclose even when no harm has happened and even when the influence is only apparent. Theory of conflict of interest (actual versus potential, situations, cases) is in section t21.

Full disclosure (slide 21)

"An engineer shall not accept financial or other compensation from **more than one party for services rendered on one project** unless the details are **fully disclosed and agreed by all parties**."

Being paid twice for one project creates divided loyalty. It is allowed only if two things happen: full disclosure of the details and the agreement of every party. Do not confuse this "full disclosure" with the society-group "duty of full disclosure" (rule 6, about naming the party you speak for). The words are the same but the rules are different.

Government conflicts of interest (slide 22)

"To avoid conflicts of interest, engineers shall **not solicit or accept a professional contract from a governmental body on which a principal or officer of their firm serves** as a member. An engineer who is a principal or employee of a private firm and who serves as a member of a governmental body **shall not participate in decisions** relating to the professional services solicited or provided by the firm to the governmental body."

There are two limbs. First, the firm must not seek or take a contract from a government body where one of the firm's principals or officers is a member. Second, an engineer who sits on a government body while working for a private firm must stay out of any decision about that firm's services to that body. Example (beyond the slides): an engineer who sits on a district tender board must leave the room when the board considers a bid from his own firm.

Slide heading	Core rule	The key escape or condition
Professional competence	Do not take or seal work you are not qualified or competent for	Seal only work prepared under your direct control and supervision
Validity of approvals	May coordinate a whole project	Each component signed or sealed by the engineer responsible for it
Confidentiality	Do not reveal professional information	Prior consent of employer or client, or authorized or required by law
Conflict of interest	Do not solicit or accept considerations from contractors; disclose potential conflicts	Disclosure covers influence that could, or appears to, affect judgment or service quality
Full disclosure	No pay from more than one party on one project	Fully disclosed and agreed by all parties
Government conflicts of interest	No contract from a government body your firm's principal or officer sits on	Do not take part in decisions on your firm's services

Q: **QB B4(ii) · 3 marks** **Describe a registered engineer's obligation to the following stakeholder group: (ii) Employers and clients.**

MARKER WORDS: EMPLOYERS AND CLIENTS

Say it like this:

> An engineer owes employers and clients **professional competence** (no assignments they are not qualified for), **confidentiality** (no revealing information without **prior consent**), and freedom from **conflict of interest**, which means not accepting considerations from contractors and **disclosing potential conflicts**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
professional competence	explaining why engineers refuse unqualified work	**SHOULD**
confidentiality	explaining the ban on revealing information	**MUST**
conflict of interest	explaining gifts, double pay or dual roles	**MUST**
direct or indirect considerations	explaining what may not be accepted from contractors	**SHOULD**
prior consent	explaining when confidential information may be released	**SHOULD**
direct control and supervision	explaining when an engineer may seal a design	**SHOULD**
signed or sealed	explaining the validity of approvals	**SHOULD**
fully disclosed and agreed by all parties	explaining payment from more than one party	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The six slide headings under obligation to employers and clients.
Professional competence; the validity of approvals; confidentiality requirement; conflict of interest; full disclosure; government conflicts of interest.

Professional competence, rule 1.
Engineers shall not undertake technical assignments for which they are not qualified.

Professional competence, rule 2 (the seal).
Engineers shall approve or seal only those plans or designs that deal with subjects in which they are competent and which have been prepared under their direct control and supervision.

The validity of approvals.
Engineers may coordinate an entire project provided that each design component is signed or sealed by the engineer responsible for that design component.

Confidentiality requirement.
Engineers shall not reveal professional information without the employer's or client's prior consent except as authorized or required by law.

Full disclosure (employers and clients).
An engineer shall not accept financial or other compensation from more than one party for services rendered on one project unless the details are fully disclosed and agreed by all parties.

Obligations to other engineers

Slides 23-25 cover how engineers must treat each other and the market for work. This is the group the exam calls "the engineering profession". It has three headings: obligation to potential employers, conflicts of interest, and reputations of other engineers.

Obligation to potential employers (slide 23)

"Engineers shall **not misrepresent or permit misrepresentation** of their or any of their associate's academic or professional qualifications. They shall not misrepresent their **level of responsibility or the complexity of prior assignments**. Pertinent facts relating to employers, employees, associates, joint ventures, or past accomplishments shall not be misrepresented when **soliciting employment or business**."

This is honesty in CVs and bids. Four things must not be misrepresented: qualifications (yours and your associates'), your level of responsibility, the complexity of past jobs, and facts about employers, employees, associates, joint ventures or past achievements. The words "or permit" matter: staying silent while someone else exaggerates for you is also a breach.

Conflicts of interest (slide 24)

"Engineers shall **not directly or indirectly give, solicit, or receive any gift or commission**, or other valuable consideration, **in order to obtain work**, and shall not make contribution to any political body with intent of **influencing the award of contract by governmental body**."

Work must be won on merit. Three acts are banned: giving, soliciting or receiving gifts, commissions or other valuable consideration to get work; doing so directly or indirectly; and making political contributions meant to influence a government contract. A gift to a tender official, or a "facilitation fee" to win a bid, is the typical breach (example beyond the slides).

Reputations of other engineers (slide 25)

1. "Engineers shall **not attempt to injure, maliciously or falsely**, directly or indirectly, the professional **reputations, prospects, practice or employment** of other engineers, nor **indiscriminately criticize** the work of other engineers."
2. "**Criticize cautiously and objectively** with respect to the person's professional status."

The rule does not ban criticism. It bans malicious or false injury and indiscriminate criticism. Where criticism is justified (for example, you find a design fault in a colleague's work) it must be cautious, objective, and mindful of the person's professional status. Compare rule 8 in the society group: if you know of a possible violation you must also help the state board, so criticism through the proper channel is allowed.

> **Key point.** Slides 20, 22 and 24 all deal with conflicts of interest, in three different settings. Slide 20: gifts and double pay while working for an employer or client. Slide 22: government bodies and your own firm. Slide 24: gifts, commissions and political contributions to win work. A question on "conflict of interest clauses in the model rules" wants all three, each with its wording. The deeper theory is in section t21.

Q: `QB B4(iii) · 3 marks` **Describe a registered engineer's obligation to the following stakeholder group: (iii) The engineering profession.**

Q: `CAT 1 (ethics-cat-1) Q1c · 2 marks each` **Describe the obligations of a registered engineer to each of the following stakeholder groups: (i) society, (ii) employers and clients, (iii) the engineering profession.**

Q: `2020-21 exam Q1D · marks not stated` **Describe the obligations of a registered engineer to each of the following stakeholder groups: (i) Society, (ii) Employers and Clients, (iii) The Engineering Profession.**

Q: `Slides 20, 22, 24 · practice` **State the conflict-of-interest clauses in the UIPE/ERB model rules.**

MARKER WORDS: OBLIGATIONS TO OTHER ENGINEERS AND CONFLICT-OF-INTEREST CLAUSES

Say it like this:

> Towards other engineers, an engineer must **not misrepresent** qualifications or past responsibility, must not give, solicit or receive **any gift or commission in order to obtain work**, and must not **maliciously or falsely injure the reputations** of other engineers, criticizing only **cautiously and objectively**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
reputations	explaining the duty not to injure other engineers	**MUST**
conflicts of interest	naming the slide 20, 22 and 24 clauses	**MUST**
misrepresent qualifications	explaining honesty to potential employers	**SHOULD**
level of responsibility	explaining what may not be exaggerated on a CV	**SHOULD**
gift or commission	explaining improper ways to win work	**SHOULD**
influencing the award of contract	explaining the ban on political contributions	**SHOULD**
indiscriminately criticize	explaining what the reputation rule forbids	**SHOULD**
cautiously and objectively	explaining how legitimate criticism is done	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three slide headings under obligations to other engineers.
Obligation to potential employers; conflicts of interest; reputations of other engineers.

Obligation to potential employers.
Engineers shall not misrepresent or permit misrepresentation of their or any of their associate's academic or professional qualifications, level of responsibility, or the complexity of prior assignments, when soliciting employment or business.

Conflicts of interest (obligations to other engineers).
Engineers shall not directly or indirectly give, solicit, or receive any gift or commission, or other valuable consideration, in order to obtain work, and shall not make contribution to any political body with intent of influencing the award of contract by governmental body.

Reputations of other engineers.
Engineers shall not attempt to injure, maliciously or falsely, directly or indirectly, the professional reputations, prospects, practice or employment of other engineers, nor indiscriminately criticize their work. Criticize cautiously and objectively with respect to the person's professional status.

Government conflicts of interest (second limb).
An engineer who is a principal or employee of a private firm and who serves as a member of a governmental body shall not participate in decisions relating to the professional services solicited or provided by the firm to the governmental body.

Contract law and the problems of law in engineering

~25 min to read | Medium priority | 3 questions

Source: Set 4, slides 26-29; Set 6, page 3 (second half) and page 4 (first half)

Contract law: what makes a contract

The slides place contract law under the heading "Engineering Ethics and Legal Issues". Engineers work through contracts all the time: with clients, employers, suppliers and subcontractors. So an engineer needs to know when a promise becomes legally binding. This links to the obligations to employers and clients in section t09.

The slide definition is short. A **contract** is a **mutual agreement between two or more parties to engage in a transaction which provides benefits to each of them**. Three elements are listed.

1. **Mutual consent.** All parties agree to the same thing, freely and knowingly. There is a "meeting of minds". If one side is forced or misled, there is no real consent.
2. **Offer and acceptance.** One party makes an offer with definite terms. The other accepts it. Acceptance must match the offer. If the other party changes the terms, that is a counter-offer, not an acceptance.
3. **Consideration.** Each side gives or promises something of value to the other. A contract is an exchange, not a one-way gift. A promise to give something for nothing is normally not enforceable.

> **Key point.** Three elements, in this order: mutual consent, offer and acceptance, consideration. If any one is missing, there is no legally enforceable contract.

Other contract issues

Slide 27 adds three points about when an agreement is legally enforceable.

- It needs **a definite promise by each party to do something specific**. Vague statements such as "we will look after you" are not enough. The promise must say what will be done.
- Each party must receive **some benefit it did not have before**. This is the slide's way of describing consideration.
- The agreement **does not have to be in writing to be valid**. A spoken agreement can bind. Writing matters mainly as proof, not as a requirement. `beyond the slides` Some contracts, such as sale of land, do need writing by statute, but the slides do not cover this.

Breach of contract

Slide 28: a breach is **an actual violation of the terms in the contract**. It must really occur. A fear that the other side might fail is not a breach. The slide gives three points.

1. Breach happens when items are **not supplied**, are **supplied but of substandard quality**, or are **not supplied until long after a deadline**.
2. The party in breach is **required to provide an equivalent value previously offered**. The other side must be put in the position it was promised.
3. If a party cannot fulfil the contract, it is **under an ethical and legal imperative to do everything possible to provide equivalent value** to the other party. Inability does not remove the duty. It changes it into a duty to make up the loss as far as possible.

> **Watch out.** Point 3 joins ethics and law in one sentence: the imperative is "ethical and legal". Quote both words when you discuss breach in an ethics answer.

The letter versus the spirit of the law

Slide 29 gives one instruction: **"read between the lines"** in terms of the **intent** of the documents **as understood by those who formulated them**. The letter of a law or contract is its exact words. The spirit is the purpose its authors had. An ethical engineer follows the purpose, not only the wording. Someone who obeys the letter and defeats the spirit is acting like the "minimal compliance" described below. The difference between law and ethics is covered in section t06.

MARKER WORDS: CONTRACT LAW

Say it like this:

> A contract is a **mutual agreement between two or more parties** to engage in a transaction which **provides benefits to each of them**. Its elements are **mutual consent**, **offer and acceptance** and **consideration**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
definite promise by each party to do something specific	saying when an agreement is enforceable	**SHOULD**
some benefit received that each did not have before	explaining consideration	**SHOULD**
does not have to be in writing	discussing oral agreements	**SHOULD**
actual violation of the terms	defining breach	**SHOULD**
substandard quality, long after a deadline	giving examples of breach	**SHOULD**
equivalent value	saying what a party in breach must provide	**SHOULD**
ethical and legal imperative	explaining the duty when a party cannot perform	**SHOULD**
read between the lines, intent	explaining letter versus spirit	**SHOULD**

Tag note: no past paper or solution file uses these phrases, so all are Should (slides only).

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define contract law (slide wording).
Mutual agreement between two or more parties to engage in a transaction which provides benefits to each of them.

The three elements of a contract.
1. Mutual consent. 2. Offer and acceptance. 3. Consideration.

Three conditions for a legally enforceable agreement (Other Contract Issues).
A definite promise by each party to do something specific; some benefit received that each did not have before; it does not have to be in writing to be valid.

Define breach of contract.
An actual violation of the terms in the contract.

Forms of breach of contract.
Items not supplied; supplied but of substandard quality; not supplied until long after a deadline.

What does "the letter vs. the spirit of the law" ask you to do?
"Read between the lines" in terms of the intent of those documents as understood by those who formulated them.

Case question: Madge and Harold

This is the contract question in the bank. Read it as a law problem first, then as an ethics problem.

Q: QB C1 · 8 marks **Madge's shop has been robbed. She says to Harold, a police officer: "If you find the thief, I will pay you $500." Harold, through the course of his police investigations, finds the thief, and the thief is convicted of the robbery. Madge refuses to pay. Advise Harold. [reconstructed from garbled scan]**

MARKER WORDS: THE MADGE AND HAROLD ANSWER

Say it like this:

> Madge's promise is an **offer** but there is no **consideration**, because Harold gave no **benefit he did not have before**: finding thieves is his **existing public duty**. So the promise is probably not **legally enforceable**, but refusing to pay breaches the **spirit** and is unethical.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
offer and acceptance	testing formation	**SHOULD**
consideration	naming the crux of the case	**SHOULD**
benefit that each did not have before	showing why police duty fails	**SHOULD**
definite promise	showing the offer was specific	**SHOULD**
not in writing	dismissing the "it was only spoken" objection	**SHOULD**
letter versus spirit	moving from law to ethics	**SHOULD**

▶ **Recall drill: complete the one-line advice, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The Madge and Harold verdict on the three contract elements.
Offer and acceptance: yes; mutual consent: yes; consideration: no (Harold was already bound by his police duty), so no binding contract.

The problems of law in engineering

Set 6 lists five problems. The first is called the greatest. Learn all five, because the question asks for any three.

1. **Minimal compliance.** This is the **greatest problem**. Engineers and employers search for **loopholes** in the law, to **barely keep to its letter while violating its spirit**. Engineers also tend to copy **standard readymade specifications** instead of coming up with innovative ideas, because meeting the standard is enough to be safe from blame. Minimal compliance led to the tragedy of the **Titanic**.
   The Titanic. beyond the slides The ship met the regulations of its day, which set the number of lifeboats by the ship's tonnage and not by the number of passengers. It therefore carried far fewer lifeboats than people on board. The law was obeyed and over 1,500 people still died. Set 6 itself

describes the disaster as an emergency (collision with an iceberg at night) plus unpreparedness: too few lifeboats, inadequate training, and warnings of icebergs unheeded.

2. **Constantly updated laws lag behind technology.** Continually updating laws and regulations can be **counter-productive**. It makes the **law always lag behind technology**, and it **overburdens the rules and the regulators**.
3. **"Non-laws".** Many laws are laws **without enforceable sanctions**. They serve only as **window dressing** and give the public a **false sense of security**.
4. **Burdening the law on purpose.** Opponents of a law may deliberately load it with **many unreasonable provisions**, so that its **repeal** is not far off.
5. **Powerful organizations break the law.** Very powerful bodies, such as the **government**, may **violate the laws when they think they can get away with it**, by inviting challengers to face them in **lengthy and costly court proceedings**. This creates **frustration with the law**.

Key point. For a "list any three" question, give minimal compliance first (with Titanic), then two more. Name each in a short phrase and add one sentence of explanation. The marks go to the named problems.

Q: **2017 Q4E · 5 marks** **List any three problems associated with application of the law in the field of Engineering.**

MARKER WORDS: PROBLEMS OF LAW IN ENGINEERING

Say it like this:

> The greatest problem of law in engineering is **minimal compliance**: engineers and employers search for **loopholes** to **barely keep to the letter of the law while violating its spirit**, as in the **Titanic** tragedy.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
minimal compliance	naming the greatest problem	**SHOULD**
loopholes, letter and spirit	explaining minimal compliance	**SHOULD**
standard readymade specifications	showing minimal compliance blocks innovation	**SHOULD**
law lags behind technology	criticising constant updating	**SHOULD**
non-laws, no enforceable sanctions	discussing weak legislation	**SHOULD**
window dressing, false sense of security	saying what non-laws do to the public	**SHOULD**
lengthy and costly court proceedings	explaining how powerful bodies evade the law	**SHOULD**

Tag note: grep over the past papers and solution files finds the question wording "application of the law" but none of these phrases, so all are Should.

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The five problems of law in engineering.
1. Minimal compliance. 2. Continually updating laws lag behind technology and overburden rules and regulators. 3. "Non-laws" with no enforceable sanctions. 4. Opponents burden the law with unreasonable provisions so repeal follows. 5. Powerful organizations violate laws and use lengthy, costly court proceedings.

What is "minimal compliance"?
Searching for loopholes in the law to barely keep to its letter while violating its spirit.

What are "non-laws"?
Laws without enforceable sanctions; they serve as window dressing and give the public a false sense of security.

The role of law in engineering

After the problems, Set 6 insists that **it is wrong to write off rule-making and rule-following as futile**. **Good laws, effectively enforced, clearly produce benefits.** The slides give these roles.

- Laws ensure **reasonable minimum standards of professional conduct**.
- They give most people and corporations a **self-interested motive to comply**, because breaking the law costs them.
- They are **powerful support and defence for those who wish to act ethically** in situations where ethical conduct might not be welcome. An engineer who refuses an unsafe shortcut can point to the law. Codes of ethics play a similar part (section t11).
- Viewing **engineering as social experimentation** gives engineers a better perspective on laws and regulations.

Precise rules versus broad guidelines

The slides say which kind of rule fits which situation.

Situation	Kind of rule that fits
Misconduct that violates **well established and regularly re-examined procedures** whose purpose is the **safety of the public**	**Precise rules and enforceable sanctions**
Areas of **experimentation**, where outcomes cannot be fully known	**Broad based guidelines** that **hold the engineer accountable** for his or her decisions. Rules must not try to cover all possible outcomes of an experiment, nor force a rigidly specified course of action.

The reason is practical. Where the procedure is settled and safety is at stake, a fixed rule is easy to check and punish. Where the work is new, a rigid rule would block good judgement or would be out of date quickly, which is problem 2 above.

Q: **Practice · 4 marks** **State two benefits of law to the engineering profession and say when precise rules suit better than broad guidelines. (Original practice question, not from the bank.)**

MARKER WORDS: THE ROLE OF LAW

Say it like this:

> Good laws, effectively enforced, ensure **reasonable minimum standards** of professional conduct. **Precise rules and enforceable sanctions** suit well established procedures for public safety, while **broad based guidelines** that hold the engineer **accountable** suit experimentation.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
reasonable minimum standards	stating what law achieves	**SHOULD**
self-interested motive to comply	explaining why people obey	**SHOULD**
support and defence for those who act ethically	linking law to ethical engineers	**SHOULD**
social experimentation	giving the perspective on laws	**SHOULD**
precise rules, enforceable sanctions	public safety procedures	**SHOULD**
broad based guidelines	regulating experimentation	**SHOULD**

Tag note: none of these phrases appears in the past papers or solution files, so all are Should.

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Roles of law in engineering (Set 6).
Ensures reasonable minimum standards of professional conduct; gives a self-interested motive to comply; powerful support and defence for those who wish to act ethically; viewing engineering as social experimentation gives a better perspective on laws.

When are precise rules and enforceable sanctions appropriate?
In cases of misconduct that violate well established and regularly re-examined procedures whose purpose is the safety of the public.

What kind of regulation fits areas of experimentation?
Broad based guidelines that hold the engineer accountable for his or her decisions; rules must not cover all possible outcomes nor force a rigidly specified course of action.

Codes of ethics: functions, limitations and canons

~55 min to read | High priority | 16 questions

Source: Set 6, pages 1-3 (ABET principles and canons, functions, limitations); Set 4, slides 30-32 (ASME principles and canons); Set 2, slides 30, 31 and 35 (three components, NSPE canons, notes about the code). The "problems of law" and "role of law" material on Set 6 pages 3-4 is in section t10. How the model rules apply to society, employers and the profession is in section t09.

What a code of ethics is, and what it contains

A code of ethics is a written statement published by a professional society that sets out the standards of conduct its members are expected to keep. The slides never give a one-line definition, so the sentence below is general knowledge. **beyond the slides** Everything after it is from the slides.

The Set 2 slides open this topic by pointing out that engineering has many fields (civil, mechanical, electrical, software, industrial) and that the same ethical issues arise across all of them. So one code of ethics pertains to engineers of all kinds.

The three components of the Engineering Code of Ethics

1. **The Fundamental Canons.** They articulate the basic components of ethical engineering.
2. **The Rules of Practice.** They clarify and specify in detail the fundamental canons of ethics in engineering.
3. **Professional Obligations.** They elaborate the obligations that engineers have.

So the canons are the short headline statements. The rules of practice spell each one out. The obligations (to society, employers and clients, and the profession) are covered in t09.

Important notes about the code (Set 2, slide 35)

- It is **not a legally binding document**.
- It is not something we want (or need) engineers to **memorize**.
- It is something we want engineers to **understand and be able to live by** as engineers.
- However, in the beginning, knowing the code is a guide to understanding how to apply it.

> **Watch out.** The slide says the code need not be memorized, but your CAT asks you to list canons and principles. For the exam you must reproduce the wording. Learn it by understanding each line, then use the drills below.

Why engineers should follow the code (Set 2 clicker question)

The slide asks why engineers should follow their professional code and offers four options. The answer the slide is steering towards is the one in the middle: the code provides a clear definition of what the public has a right to expect from responsible engineers. The other options are weaker reasons. The public will trust engineers more if they know there is a code (true, but a by-product). It helps avoid legal problems (the code

is not a law). It raises the image of the profession and so gets engineers more pay (a motive, not an ethical reason). **beyond the slides** The slide does not print the answer; this reading follows the notes on slide 35.

Q: **2016 Q2A · 4 marks** **What is a professional code of ethics?**

Q: **2017 Q3A · 4 marks · see also t21** **Describe what is meant by a Code of Ethics citing clear supporting examples.**

Q: **2020-21 Q1A, QB B2 · 5 marks** **What is meant by a Code of Ethics (COE) and what are the known weaknesses of CoEs?**

Q: **2017 Q3B · 6 marks** **Give four (4) reasons behind why these codes are important to society.**

Q: **2015 Q1B · 2 marks · 2011/12** **What is the purpose of a Code of Ethics in engineering practice?**

Q: **CAT 2021-22 Q1A (as worded in the solution file)** **What do you understand by a professional code of conduct and how is it important for any engineer?**

FLASHCARDS: CLICK TO FLIP

The three components of the Engineering Code of Ethics.
The Fundamental Canons; the Rules of Practice; Professional Obligations.

What do the Fundamental Canons do?
They articulate the basic components of ethical engineering.

What do the Rules of Practice do?
They clarify and specify in detail the fundamental canons of ethics in engineering.

What do the Professional Obligations do?
They elaborate the obligations that engineers have.

Four notes about the code of ethics (Set 2).
It is not a legally binding document; it is not something we want (or need) engineers to memorize; it is something we want engineers to understand and be able to live by; in the beginning, knowing the code is a guide to understanding how to apply it.

MARKER WORDS: THE CODE AND ITS THREE COMPONENTS

Say it like this:

> The Engineering Code of Ethics has **three components**: the **fundamental canons**, which articulate the basic components of ethical engineering; the **rules of practice**, which clarify and specify the canons in detail; and **professional obligations**. It is **not a legally binding document**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
fundamental canons	describing what the code contains, or listing the canons	**MUST**
professional obligations	naming the third component of the code	**MUST**
rules of practice	saying how the code spells out the canons	**SHOULD**
articulate the basic components of ethical engineering	defining what a canon is	**SHOULD**
not legally binding	contrasting a code with the law	**SHOULD**
understand and live by	explaining why the code is not for memorizing	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

The fundamental principles: ABET and ASME

The slides give two codes with the same opening. ABET (Accreditation Board for Engineering and Technology) is on the Set 6 notes. ASME is on Set 4, slide 30. Both begin: "Engineers shall uphold and advance the integrity, honor, and dignity of the engineering profession by:" and then list the principles. ABET has four. ASME has three. The ASME slide uses "Engineers uphold and advance", without "shall".

#	ABET Fundamental Principles (4)	ASME Fundamental Principles (3)
1	using their knowledge and skill for the enhancement of the **human race**	using their knowledge and skill for the enhancement of **human welfare**
2	being honest and impartial and serving with fidelity the public, their employers, and clients	being honest and impartial, and serving with fidelity the public, their employers and clients
3	striving to increase the competence and prestige of the engineering profession	striving to increase the competence and prestige of the engineering profession
4	supporting the professional and technical societies of their discipline	not in the ASME list

Key point. The only wording difference is "human race" (ABET) against "human welfare" (ASME). The only extra principle is the fourth, ABET's "supporting the professional and technical societies of their discipline". For a 4-mark "list four fundamental principles" answer, give the ABET four.

Read as a set, the four principles say: use your skill to benefit people; be honest, impartial and loyal to the public, employers and clients; raise the standing of the profession; back your professional societies.

Q: **2016 Q2D · 4 marks · 2017 Q4C · CAT 2021-22 Q1C** **List four fundamental principles of the Engineers' code of ethics (common to the different engineering society codes of ethics).**

Q: **CAT 1 2015 Q2C · 5 marks** **Highlight any 5 points of the IEEE code of ethics.**

FLASHCARDS: CLICK TO FLIP

The four ABET fundamental principles (after the lead-in "Engineers shall uphold and advance the integrity, honor, and dignity of the engineering profession by").
Using their knowledge and skill for the enhancement of the human race; being honest and impartial and serving with fidelity the public, their employers, and clients; striving to increase the competence and prestige of the engineering profession; supporting the professional and technical societies of their discipline.

The three ASME fundamental principles.
Using their knowledge and skill for the enhancement of human welfare; being honest and impartial, and serving with fidelity the public, their employers and clients; striving to increase the competence and prestige of the engineering profession.

The lead-in sentence shared by the ABET and ASME principles.
Engineers shall uphold and advance the integrity, honor, and dignity of the engineering profession by:

MARKER WORDS: THE FUNDAMENTAL PRINCIPLES

Say it like this:

> Engineers shall **uphold and advance the integrity, honor, and dignity** of the profession by using their **knowledge and skill for the enhancement of the human race**, being **honest and impartial** and serving with **fidelity** the public, employers and clients, striving to increase its **competence and prestige**, and **supporting the professional and technical societies** of their discipline.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
integrity, honor, and dignity	opening any answer on the principles	**SHOULD**
honest and impartial	giving the second principle	**SHOULD**
serving with fidelity	saying who engineers owe loyalty to: public, employers, clients	**SHOULD**
competence and prestige	giving the third principle	**SHOULD**
enhancement of the human race / human welfare	giving the first principle	**SHOULD**
professional and technical societies	giving the fourth (ABET only) principle	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

The fundamental canons: ABET, ASME and NSPE

The slides carry three lists of canons. ABET (Set 6) has **seven**. ASME (Set 4, slides 31 and 32, split over two slides) has **seven**. NSPE (Set 2, slide 31) has **six**. Set 6 spells the heading "Fundamental Cannons"; the correct spelling is canons, and that is what you should write.

> **Key point.** When the exam says "list the seven fundamental canons", give the ABET list below in this order. It includes the honor, integrity and dignity canon that most seven-canon lists use. The ASME seven (below) is also a valid slide list of exactly seven.

The seven canons, in slide wording (ABET, Set 6 page 1)

Engineers shall:

1. hold paramount the safety, health, and welfare of the public in the performance of their professional duties;
2. perform service only in areas of their competence;
3. issue public statements only in an objective and truthful manner;
4. act in professional matters for each employer or client as faithful agents or trustees, and shall avoid conflicts of interest;
5. build their professional reputations on the merits of their services and shall not compete unfairly with others;
6. act in such manner as to uphold and enhance the honor, integrity and dignity of the profession;
7. continue their professional development throughout their careers, and shall provide opportunities for the professional development of those engineers under their supervision.

One memory hook, in order: **public, competence, statements, employer, reputation, honor, development**.

The ASME canons (Set 4, slides 31 and 32)

Slide 31 gives three and slide 32 gives four. The ASME list reorders the ABET canons and swaps one: it has no "honor, integrity and dignity" canon and adds "associate only with reputable persons or organizations".

1. Engineers shall hold paramount the safety, health and welfare of the public in the performance of their professional duties.
2. Engineers shall perform services only in the areas of their competence.
3. Engineers shall continue their professional development throughout their careers and shall provide opportunities for the professional development of those engineers under their supervision.
4. Engineers shall act in professional matters for each employer or client as faithful agents or trustees, and shall avoid conflicts of interest.
5. Engineers shall build their professional reputations on the merit of their services and shall not compete unfairly with others.
6. Engineers shall associate only with reputable persons or organizations.
7. Engineers shall issue public statements only in an objective and truthful manner.

The NSPE canons (Set 2, slide 31)

"Engineers, in the fulfillment of their professional duties, shall:"

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully, so as to enhance the honor, reputation, and usefulness of the profession.

The Set 2 bolts exercise (a sub-contractor supplies foreign-made bolts where only U.S.-made parts are allowed) ties the answer "tell the customer" to two NSPE canons: "Avoid deceptive acts" and "Act for each employer or client as faithful agents or trustees". Full retelling is in t04.

Side by side: where each wording appears

Subject	ABET (7)	ASME (7)	NSPE (6)
Public safety	Hold paramount the safety, health, and welfare of the public in the performance of their professional duties	same wording	Hold paramount the safety, health, and welfare of the public
Competence	Perform service only in areas of their competence	Perform services only in the areas of their competence	Perform services only in areas of their competence
Public statements	Issue public statements only in an objective and truthful manner	same	same
Employer or client	Act in professional matters for each employer or client as faithful agents or trustees, and shall avoid conflicts of interest	same	Act for each employer or client as faithful agents or trustees (no conflicts clause)
Reputation, fair competition	Build their professional reputations on the merits of their services and shall not compete unfairly with others	same, "the merit" singular	not present
Honor of the profession	Act in such manner as to uphold and enhance the honor, integrity and dignity of the profession	not present	Conduct themselves honorably, responsibly, ethically, and lawfully, so as to enhance the honor, reputation, and usefulness of the profession
Professional development	Continue their professional development throughout their careers, and provide opportunities for those under their supervision	same	not present
Reputable associates	not present	Associate only with reputable persons or organizations	not present
Deceptive acts	not present	not present	Avoid deceptive acts

The same comparison for the principles is in the table in the previous topic. In one line: the ABET and ASME principles share three items and ABET adds the societies principle; the ABET and ASME canons share six items; NSPE shares the first four (without the conflicts clause) and adds "avoid deceptive acts".

Watch out. The unofficial solution files give a different "seven canons": the six NSPE canons plus "Be fair and supportive to their colleagues, and promote professional growth and opportunities for professional development". That seventh line is not on any of your slides. The solution file for the UIPE question gives only five, the first with an added phrase "and strive to comply with the principles of sustainable development" (which is the wording in the UIPE/NSPE revised code, **beyond the slides**). The slides win: give the ABET seven. If the marker is working from the solution file, your list will still match canons 1 to 4 and the competence, honor and development ideas, which is where the marks are.

Q: **QB B3 · 7 marks · 2020-21 Q1B · 2016 Q3B** **List the fundamental canons of the UIPE (Uganda Institute of Professional Engineers) Code of Ethics. Hint: similar to IEEE fundamental canons. (List 7 fundamental canons within the codes of ethics.)**

Q: **2013/14 paper · 8 marks [reconstructed from garbled scan]** **What are the common features contained in a code of ethics for engineers?**

FLASHCARDS: CLICK TO FLIP

The seven fundamental canons (ABET, Set 6). Start: "Engineers shall..."
Hold paramount the safety, health, and welfare of the public in the performance of their professional duties; perform service only in areas of their competence; issue public statements only in an objective and truthful manner; act in professional matters for each employer or client as faithful agents or trustees, and shall avoid conflicts of interest; build their professional reputations on the merits of their services and shall not compete unfairly with others; act in such manner as to uphold and enhance the honor, integrity and dignity of the profession; continue their professional development throughout their careers, and shall provide opportunities for the professional development of those engineers under their supervision.

The six NSPE fundamental canons.
Hold paramount the safety, health, and welfare of the public; perform services only in areas of their competence; issue public statements only in an objective and truthful manner; act for each employer or client as faithful agents or trustees; avoid deceptive acts; conduct themselves honorably, responsibly, ethically, and lawfully, so as to enhance the honor, reputation, and usefulness of the profession.

The seven ASME fundamental canons.
Hold paramount the safety, health and welfare of the public in the performance of their professional duties; perform services only in the areas of their competence; continue their professional development throughout their careers and provide opportunities for those under their supervision; act as faithful agents or trustees for each employer or client and avoid conflicts of interest; build their professional reputations on the merit of their services and not compete unfairly; associate only with reputable persons or organizations; issue public statements only in an objective and truthful manner.

Memory hook for the seven ABET canons, in order.
Public, competence, statements, employer, reputation, honor, development.

Which canon is in ASME only?
Engineers shall associate only with reputable persons or organizations.

Which canon is in NSPE only?
Avoid deceptive acts.

MARKER WORDS: THE FUNDAMENTAL CANONS

Say it like this:

> Engineers shall **hold paramount the safety, health, and welfare of the public**, **perform service only in areas of their competence**, **issue public statements only in an objective and truthful manner**, act as **faithful agents or trustees** and **avoid conflicts of interest**, build reputations on merit without **competing unfairly**, **uphold and enhance the honor, integrity and dignity** of the profession, and **continue their professional development**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
hold paramount the safety, health, and welfare of the public	giving canon 1, or arguing that public safety outranks employer loyalty	**MUST**
perform services only in areas of their competence	giving canon 2, or discussing the principle of individual competence	**MUST**
objective and truthful manner	discussing public statements, reports, or whistleblowing	**MUST**
faithful agents or trustees	explaining duties to an employer or client	**MUST**
avoid conflicts of interest	closing the employer/client canon	**MUST**
avoid deceptive acts	giving the NSPE canon, or any honesty case such as the bolts exercise	**MUST**
professional development	giving the last ABET canon	**MUST**
not compete unfairly	giving canon 5	**SHOULD**
associate only with reputable persons or organizations	giving the ASME-only canon	**SHOULD**

▶ **Recall drill: write the first four canons from memory, then click each blank to check**

The seven functions of codes of ethics

Set 6 lists seven roles or functions. Six are the useful functions. The seventh is listed as a function but is actually a warning. In order:

1. Inspiration and guidance

- Codes provide a **positive stimulus** for ethical conduct and helpful guidance by using **positive language**.
- Codes should be **brief** to be effective, so they offer only **general guidance**.
- A number of societies or professional bodies add **supplementary statements or guidelines** giving specific directions.

2. Support

- Codes give **positive support** to those seeking to act ethically.
- An engineer under pressure to act unethically can use one of the **publicly proclaimed codes** to get support for his stand on specific moral issues.
- Codes also serve as **legal support** for engineers.

3. Deterrence and discipline

- Codes can be used as a **basis for conducting investigations** on unethical conduct.
- They also provide a **deterrent** for engineers to act immorally.
- Engineers punished by professional societies for proven unethical behaviour, by **revoking the right to practise as engineers**, also suffer **public ridicule and loss of respect** from colleagues and the local community.
- This helps produce ethical conduct, even though it can be viewed as a **negative way of motivation**.

4. Education and mutual understanding

Codes can be used for **discussion and reflection** on moral issues, and so improve the understanding of moral responsibilities among all engineers, clients, the public and good organizations.

5. Contributing to the profession's public image

Codes present the engineering profession as an **ethically committed** society in the eyes of the public, enhancing its image.

6. Protecting status quo

Codes establish **ethical conventions**, which can help promote an **agreed upon minimum level** of ethical conduct.

7. Promoting business interests

Codes can place **unwarranted restraints of commerce** on business dealings. This is a danger, not a benefit: a code written to protect members' business can limit honest competition.

Watch out. Functions 5 (preserving the profession's public image) and 6 (protecting the status quo) are the two the slide says have been abused: the worst abuse is using "preserving profession's public image" and "protecting status quo" to restrict honest moral effort. Function 7 is listed as a function but is itself a warning (unwarranted restraints of commerce). If a question says "list six roles", give 1 to 6 and say that 5 and 6 can be abused. Add 7 as a negative if asked for more.

Q: **2016 Q2B · 10 marks · 2011/12** **List any SIX roles/functions of professional codes of ethics.**

Q: **Practice · 4 marks** **Explain how a code of ethics supports an engineer, and how it deters misconduct.**

FLASHCARDS: CLICK TO FLIP

The seven functions of codes of ethics (Set 6).
1 Inspiration and guidance; 2 Support; 3 Deterrence and discipline; 4 Education and mutual understanding; 5 Contributing to the profession's public image; 6 Protecting status quo; 7 Promoting business interests.

Function 6: protecting status quo.
Codes establish ethical conventions, which can help promote an agreed upon minimum level of ethical conduct.

Function 7: promoting business interests.
Codes can place unwarranted restraints of commerce on business dealings.

Function 5: contributing to the profession's public image.
Codes present the engineering profession as an ethically committed society in the eyes of the public, thus enhancing their image.

Function 4: education and mutual understanding.
The codes can be used for discussion and reflection on moral issues and thereby improve the understanding of moral responsibilities among all engineers, clients, public and good organizations.

The relative importance of the functions

Set 6 ranks the functions through the idea of **engineering as social experimentation**: a new project is an experiment on society, so the people running it must be free to say what the risks are.

- **Support is the primary function.** Only this support enables engineers to speak out clearly and openly their views to those affected by engineering projects.

- **Inspiration and guidance** and **education** are also important, in promoting mutual understanding and in motivating engineers to act with higher moral standards.
- **Discipline is of secondary importance.** People with unethical conduct, when exposed, are subject to law. Building elaborate paralegal procedures inside professional societies duplicates a job the legal system does better. At best, codes should discipline engineers in areas not covered by law.
- **The worst abuse of codes** has been to restrict honest moral effort in the name of "preserving the profession's public image" and "protecting status quo". The best way to increase trust is by encouraging and aiding engineers to speak freely and responsibly about public safety.

So the ranking is: support first; inspiration, guidance and education next; discipline second-tier; and the image and status quo functions are the ones most open to abuse. The link to the law is developed in t10.

Q: **Practice · 5 marks** **Which function of codes of ethics is the most important, and which is the most open to abuse? Give reasons.**

MARKER WORDS: FUNCTIONS OF CODES

Say it like this:

> Codes of ethics serve seven functions: **inspiration and guidance**, **support**, **deterrence and discipline**, **education and mutual understanding**, contributing to the profession's **public image**, **protecting status quo** and **promoting business interests**. The **primary function is support**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
inspiration and guidance	naming function 1	**SHOULD**
publicly proclaimed codes	explaining how a code supports an engineer under pressure	**SHOULD**
basis for investigations and **deterrent**	explaining deterrence and discipline	**SHOULD**
mutual understanding	naming function 4	**SHOULD**
ethically committed	explaining the public image function	**SHOULD**
agreed upon minimum level	explaining protecting status quo	**SHOULD**
unwarranted restraints of commerce	explaining promoting business interests as a danger	**SHOULD**
engineering as social experimentation	justifying why support is the primary function	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

The four limitations of codes of ethics

1. General and vague wording

Codes are restricted to **general and vague wording**. They cannot be straightaway applied to all situations. It is impossible to foresee the full range of moral problems that can arise in a complex profession like engineering. Example: "hold paramount the safety of the public" does not tell you how much safety margin a given bridge needs.

2. Clauses can conflict

It is easy for **different clauses to come into conflict** with each other. Usually codes provide no guidance as to which clause should have priority, so they create **moral dilemmas**. Example: "act as faithful agents" to an employer against "hold paramount the safety of the public" when the employer wants a defect hidden.

3. Not the final moral authority

Codes **cannot serve as the final moral authority** for professional conduct. If a professional society's code is taken as the last word, we have adopted one particular set of conventions, that is **ethical conventionalism**. Ethical conventionalism is the view that what is right is simply what the accepted conventions say.

4. Separate codes make ethics look relative

Andrew Oldenquist and Edward Slowter pointed out that **separate codes for different professional societies** can give members the feeling that ethical conduct is **more relative than it is**, and can convey to the public the view that **none is "really right"**. The slides close this limitation with a balanced line: the current codes are by no means perfect but are definitely steps in the right direction.

> **Key point.** Always add the closing line. It shows the marker you are weighing limitations, not dismissing codes: "not perfect, but steps in the right direction".

> **Q:** **2016 Q2C · 6 marks · 2017 Q3C · 4 marks** **Describe any FOUR limitations of a professional code of ethics. (2017: Discuss two limitations associated with the use/application of professional codes or codes of conduct.)**

> **Q:** **CAT 2021-22 Q1D** **Discuss four limitations of the use of professional codes of conduct and use one of the ethical theories either to support or refute these limitations.**

FLASHCARDS: CLICK TO FLIP

The four limitations of codes of ethics (Set 6), as headings.
1 Restricted to general and vague wording; 2 Different clauses can come into conflict, with no guidance on priority, creating moral dilemmas; 3 They cannot serve as the final moral authority (ethical conventionalism); 4 Separate codes for different societies can make ethical conduct seem more relative than it is (Oldenquist and Slowter).

Ethical conventionalism (in the context of codes).
Taking the code of a professional society as the last word, so we are getting into a particular set of conventions.

Who pointed out that separate codes make ethics seem relative, and what did they say?
Andrew Oldenquist and Edward Slowter: separate codes for different professional societies can give members the feeling that ethical conduct is more relative than it is, and can convey to the public the view that none is "really right".

MARKER WORDS: LIMITATIONS OF CODES

Say it like this:

> Codes of ethics are limited because they use **general and vague wording**, their **clauses can conflict** without a priority rule, they **cannot serve as the final moral authority** (**ethical conventionalism**), and **separate codes** make ethics seem **more relative than it is**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
general and vague wording	giving limitation 1	**SHOULD**
clauses come into conflict	giving limitation 2	**SHOULD**
moral dilemmas	saying what conflicting clauses create	**SHOULD**
final moral authority	giving limitation 3	**SHOULD**
ethical conventionalism	naming the risk of treating the code as the last word	**SHOULD**
more relative than it is	giving limitation 4	**SHOULD**
steps in the right direction	closing any limitations answer	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

Writing a code for a small high-tech firm

The 2014/15 paper (Q1d, **7 marks**) asks you to act as president of a small high-technology firm that has grown to the point where employees need ethics guidelines. You must first decide what type of code suits the firm, then write a code that includes the features asked for. The slides give no types of code, so the "types" part below is **beyond the slides**. Build the code itself from the slide canons and the three components of the code.

Q: **2015 Q1D · 7 marks** **Imagine you are the president of a small high technology firm. Your company has grown over the last years to the point where you feel that it is important that your employees have some guidelines regarding ethics. Start by deciding what type of code is appropriate for your company. Then finally write a code that incorporates these features.**

FLASHCARDS: CLICK TO FLIP

The three components of a code, as a checklist for writing your own code.
Fundamental canons; rules of practice; professional obligations.

Two statements about codes that are brief (Set 6, function 1).
Codes should be brief to be effective and hence such codes offer only general guidance; supplementary statements or guidelines add specific directions.

MARKER WORDS: WRITING AND USING A CODE

Say it like this:

> A good code is **brief** and uses **positive language** to give **inspiration and guidance**; it states **fundamental canons**, adds **supplementary guidelines** for specific directions, and supports engineers who **act ethically** under pressure.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
brief to be effective	explaining why codes are general	**SHOULD**
positive language	describing the inspiration function	**SHOULD**
supplementary guidelines	saying how societies give specific directions	**SHOULD**
minimum level of ethical conduct	saying what a code guarantees	**SHOULD**
conflicts of interest	writing any employer or client clause	**MUST**

▶ **Recall drill: write the sentence from memory, then click each blank to check**